CBC Formula Page 1

4 cbc formula

a) Mean Cell Volume (MCV)

15 form [a1]=([b1]/[c1])*10

a1= Mean Cell Volume (MCV)
b1= Packed Cell Volume (HCT)
c1= R.B.C Count

[Mean Cell Volume (MCV)]=([Packed Cell Volume (HCT)]/[R.B.C Count])*10

a1 = b1 / c1 *10

b) Mean Cell Haemoglobin (MCH)

15 form [a1]=([b1]/[c1])*10

a1= Mean Cell Haemoglobin (MCH)
b1= Hemoglobin (Hb)
c1= R.B.C Count

[Mean Cell Haemoglobin (MCH)]=([Hemoglobin (Hb)]/[R.B.C Count])*10

a1 = b1 / c1 *10

c) Mean Corpuscular Hb Concn. (MCHC)

11 form [a1]=([b1]/[c1])*100

a1= Mean Corpuscular Hb Concn. (MCHC)
b1= Hemoglobin (Hb)
c1= Packed Cell Volume (HCT)

[Mean Corpuscular Hb Concn. (MCHC)]=([Hemoglobin (Hb)]/[Packed Cell Volume (HCT)])*100

a1 = b1 / c1 *100

d) Differential % WBCs count

16 form [a1]=([b1]+[c1]+[d1]+[e1])/100

a1= Neutrophil
b1= Lymphocytes
c1= Eosinophil
d1= Monocytes
e1= Basophils

[Neutrophil]=([Lymphocytes]+[Eosinophil]+[Monocytes]+[Basophils])/100

e) Absolute Neutrophil Count

17 form [a1]=([b1]/[c1])*100+[neu_id]+[abs_neu_id]

a1= Absolute Neutrophil Count
b1= Total (WBC) Count
c1= Neutrophil
neu= Neutrophil
abs_Netrophil= Absolute Neutrophil Count

[Absolute Neutrophil Count]=([Total (WBC) Count]/[Neutrophil])*100+[Neutrophil]+[Absolute Neutrophil Count]

f) Absolute Lymphocyte Count

17 form [a1]=([b1]/[c1])*100+[neu_id]+[abs_neu_id]

a1= Absolute Lymphocyte Count
b1= Total (WBC) Count
c1= Lymphocytes
neu= Neutrophil
abs_Netrophil= Absolute Neutrophil Count

[Absolute Lymphocyte Count]=([Total (WBC) Count]/[Lymphocytes])*100+[Neutrophil]+[Absolute Neutrophil Count]

CBC Formula Page 2

g) Absolute Monocyte Count

17 **form** [a1]=([b1]/[c1])*100+[neu_id]+[abs_neu_id]

a1= Absolute Monocyte Count
b1= Total (WBC) Count
c1= Monocytes
neu= Neutrophil
abs_Netrophil= Absolute Neutrophil Count

[Absolute Monocyte Count]=([Total (WBC) Count]/[Monocytes])*100+[Neutrophil]+[Absolute Neutrophil Count]

h) Absolute Eosinophils Count, AEC

17 **form** [a1]=([b1]/[c1])*100+[neu_id]+[abs_neu_id]

a1= Absolute Eosinophils Count, AEC
b1= Total (WBC) Count
c1= Eosinophil
neu= Neutrophil
abs_Netrophil= Absolute Neutrophil Count

[Absolute Eosinophils Count, AEC]=([Total (WBC) Count]/[Eosinophil])*100+[Neutrophil]+[Absolute Neutrophil Count]

17 i) Absolute Basophils Count

form [a1]=([b1]/[c1])*100+[neu_id]+[abs_neu_id]

a1= Absolute Basophils Count
b1= Total (WBC) Count
c1= Basophils
neu= Neutrophil
abs_Netrophil= Absolute Neutrophil Count

[Absolute Basophils Count]=([Total (WBC) Count]/[Basophils])*100+[Neutrophil]+[Absolute Neutrophil Count]

CBC Report

Patient Name : DEMO
Age & Sex : 1.3 Years & Female
Referred By : SELF
Lab Name : SALONI DIAGNOSTICS

Reg Date : 01/05/2024 02:10 PM
Report Date : 03/05/2024 03:38 PM
Reg No : 101
Sample ID : 101

HEMATOLOGY

Complete Blood Count (CBC)

Test Description	Result	Unit	Reference Range
Hemoglobin (Hb)	**10.2**	gms/dl	10.5 - 14
R.B.C Count	**5.69**	mil./cmm	4.0 - 5.2
Packed Cell Volume (HCT)	**33.9**	%	34 - 40
Mean Cell Volume (MCV)	**59.58**	fL	75 - 87
Mean Cell Haemoglobin (MCH)	**17.93**	pg	24 - 30
Mean Corpuscular Hb Concn. (MCHC)	**30.09**	%	31 - 37
RDW-CV	**16.3**	%	11.60 - 14
Total (WBC) Count	7000	cells/c.mm	4000 - 11000
Differential % WBCs count			
Neutrophil	51	%	45 - 75
Lymphocytes	38	%	20 - 40
Eosinophil	01	%	1 - 6
Monocytes	10	%	2 - 10
Basophils	00	%	0 - 2
Absolute Differential Count			
Absolute Neutrophil Count	3570.00	/c.mm	2000 - 7000
Absolute Lymphocyte Count	2660.00	/c.mm	1000 - 3000
Absolute Monocyte Count	700.00	/c.mm	200 - 1000
Absolute Eosinophils Count, AEC	70.00	/c.mm	20 - 500
Absolute Basophils Count	0.00	/c.mm	20 - 100
Platelet Count	275000	cells/c.mm	150000 - 450000
Mean Platelet Volume (MPV)	**6.7**	fl	7.2 - 11.7
Platelet Distribution Width (PDW)	**15.8**	%	10 - 14
PCT (Platelet Crit)	0.18	%	0.17 - 0.37
Peripheral Blood Smear			
RBC Morphology	Normocytic Normochromic		
WBC Morphology	Within Normal Limits		
Platelet Morphology	Adequate on smear		
Advice	Kindly correlate clinically & close follow up.		

END OF REPORT

Mr. SUSHANT KUMAR
doctor

Page 1 of 1

Bilirubin

FORMULA

1 BILIRUBIN

[a1]=[b1]-[c1]

a1= Bilirubin Indirect
b1= Bilirubin Total (Serum)
c1= Bilirubin Direct

[Bilirubin Indirect]=[Bilirubin Total (Serum)]-[Bilirubin Direct]
a1 = b1 - c1

Bilirubin

Patient Name : DEMO
Age & Sex : 23 Years & Male
Referred By : SELF
Lab Name : SALONI DIAGNOSTICS

Reg Date : 01/05/2024 02:10 PM
Report Date : 01/05/2024 05:33 PM
Reg No : 101
Sample ID : 101

BIOCHEMISTRY

BILIRUBIN

Test Description	Result	Unit	Reference Range
Bilirubin Total (Serum)	0.9	mg/dL	0.2 - 1.2
Bilirubin Direct	0.2	mg/dL	0.0 - 0.3
Bilirubin Indirect	0.70	mg/dL	0.2 - 0.9

END OF REPORT

RFT BUL to BUN

2 RFT (BUL)BLOOD UREA LEVEL into (BUN) BLOOD UREA NITROGEN

[a1]=[b1]/2.14

a1=BLOOD UREA NITROGEN (BUN)
b1=BLOOD UREA LEVEL (BUL)

[BLOOD UREA NITROGEN (BUN)]=[BLOOD UREA LEVEL (BUL)]/2.14

a1 = b1 / 2.14

RFT BUL to BUN

Patient Name : DEMO
Age & Sex : 23 Years & Male
Referred By : SELF
Lab Name : SALONI DIAGNOSTICS

Reg Date : 01/05/2024 02:10 PM
Report Date : 03/05/2024 03:19 PM
Reg No : 101
Sample ID : 101

BIOCHEMISTRY

RENAL(KIDNEYS) FUNCTION TEST

Test Description	Result	Unit	Reference Range
BLOOD UREA LEVEL (BUL)	20	mg/dl	0 - 45
BLOOD UREA NITROGEN (BUN)	9.35	mg/dl	3.3 - 18.7

Test Information

Kidneys play several vital roles in maintaining health. One of their most important jobs is to filter waste materials from the blood and expel them from the body as urine. The kidneys also help control the levels of water and various essential minerals in the body. In addition, they're critical to the production of:

1) vitamin D
2) red blood cells
3) hormones that regulate blood pressure

Patient also need kidney function testing done if he/she has other conditions that can harm the kidneys, such as diabetes or high blood pressure.

Symptoms that may indicate a problem with your kidneys include:

1) high blood pressure
2) blood in the urine
3) frequent urges to urinate
4) difficulty beginning urination
5) painful urination
6) swelling of the hands and feet due to a buildup of fluids in the body

A single symptom may not mean something serious. However, when occurring simultaneously, these symptoms suggest that your kidneys aren't working properly. Kidney function tests can help determine the reason.

METHOD :

BLOOD UREA NITROGEN (BUN) - KINETIC UV ASSAY.
CREATININE- CREATININE ENZYMATIC METHOD
URIC ACID - URICASE / PEROXIDASE METHOD
CALCIUM - ARSENAZO III METHOD, END POINT.

END OF REPORT

this is not approved report

PT / INR

3 PT INR
form [a1]=([b1]/[c1]^[d1]
[a1]=([b1]/[cid]_[cvalue])^[did]_[dvalue]

a1= INR
b1= Patients Prothrombin Time
cid =Control Prothrombin Time
cvalue = 14 type karane
did = ISI VALUE
dvalue= 1.1 type karane

/ Prothrombin Index
= cantrol ÷ pt (Patsnt Tim
- X 100
= cantrol ÷ (pt) X 100
formula = 11

Hba1c - Formula.

Formula - (7.) [a1] = [b1]* [28.7] - [46.7] .

a1 = Average blood Glulobe [ABG].

b1 = HbA1C.

Prothrombin Ratio = $\frac{INR}{ISI}$

PT INR

Patient Name : DEMO	**Reg Date** : 01/05/2024 02:10 PM
Age & Sex : 23 Years & Male	**Report Date** : 03/05/2024 03:24 PM
Referred By : SELF	**Reg No** : 101
Lab Name : SALONI DIAGNOSTICS	**Sample ID** : 101

PT INR

Test Description	Result	Unit	Reference Range
Sample Type	Sodium Citrate		
Patients Prothrombin Time	16	seconds	12 - 16
Control Prothrombin Time	14	Seconds	-
ISI VALUE	1.1	-	-
INR	1.16	-	Standard Therapy: INR = 2 to 3 High Dose Therapy: INR = 3 to 4.5

Note ; T PT is the time taken by the liquid portion [plasma] of blood to form the clot. It is the measure of extrinsic pathway of coagulation. PT is prolonged in deficiency of clotting factors Fibrinogen I, Prothrombin II, Factor V, Factor VII and Factor X. It is also prolonged in Vit. K deficiency, liver damage, DIC, Patients on Anticoagulant Therapy and Congenital Afibrinogenemia

END OF REPORT

Lft

Albamin protin Ag Ratio

Albumin = ~~Total~~

Globumin = total protin - Albumin

fourmula - [3]

A1 = Globumin

B1 = Total protin

C1 = Albumin

RFT

Ag Ratio = Albumin ÷ Globulin

fourmula = [4]

A1 = Ag Ratio

B1 = Albumin

C1 = Globulin

- IRON - Formula.

A1 B1 C1

Saturation = IRON % TIBC x 100

∴ [A1 = B1 ÷ C1 x 100]

RFT

A1 = b1 / C1

④ Urea / Creatine ratio - Urea / Creatine
A1 B1 C1

④ BUN / Creactinine ratio - BUN / Creatinine
A1 B1 C1